

Supplementary Material: Complete Prime Factorization of 114 Neural Network Algorithms for “Computational Primes” (Bieg, 2026)

Michael Bieg
Independent Researcher
Bieg.Michael@web.de

Version 3

Legend. Pn = physical prime (analog hardware exists). Xn = missing prime (no analog equivalent).
 $\mathbf{M}$ = fully mappable to analog. $\mathbf{G}$ = GPU-mappable (X2/X4 available on GPU). $\mathbf{U}$ = unmappable (X1 or X3).

Note on this version. The table rows below are unchanged from the previous version; the “St.” column therefore carries the $v2$ status of each entry. What changed in version 3 is (i) the summary statistics at the end of this document, which did not agree with the table they summarized, and (ii) the status of the 28 non-M entries, which the main paper re-derives as $\mathbf{B}$ (bounded-mappable) or $\mathbf{D}$ (digital-by-position) in its Table “Reclassification of the 28 non-M entries”. Read the “St.” column here together with that table.

1 Attention & Transformer Core (A1–A20)

#	Algorithm	Factorization	St.	Note
A1	Scaled Dot-Product Attn	P11·P2·P3·P1·P12·P11M		Core: $QK^T/\sqrt{d}$ via P11·P2, softmax via P3·P1·P12, attn×V via P11
A2	Multi-Query Attn (MQA)	P11·P2·P3·P1·P12·P11M		Same primes as A1; shared K,V is wiring/fan-out
A3	Grouped-Query Attn (GQA)	P11·P2·P3·P1·P12·P11M		Interpolation MHA↔MQA; group = static routing
A4	FlashAttention (tiled)	P11·P2·P3·P1·P12·P5· P8	M	Online softmax adds P5 (running max) + P8 (tile accumulator); tile dispatch via X2
A5	Sliding Window Attn	P11·P2·P3·P1·P12·P7	M	P7 gates window boundary; window size is static
A6	Linear Attention	P11·P1·P12·P3	M	Kernel ϕ determines primes; ELU+1 uses P3
A7	Cross-Attention	P11·P2·P3·P1·P12·P11M		Identical to A1; cross = routing distinction
A8	Sparse Attn (BigBird)	P11·P2·P3·P1·P12·P7· P10	M	Fixed sparsity via P7 mask; random component via P10
A9	Causal Masking	P5·P7	M	Solved: race topology enforces $j \leq i$
A10	Attention Score Masking	P1·P2·P7	M	Additive $-\infty$ mask (P1·P2) or multiplicative (P7)
A11	Multi-Head Split	P2·P1	M	Parallel crossbars + concat
A12	Residual Connection	P1	M	Solved: KCL at node

#	Algorithm	Factorization	St.	Note
A13	RoPE	P11·P2	M	Fixed trig coefficients $\times$ variable
A14	Token Embedding	P8·P2	M	Crossbar with one-hot; area scales with vocab
A15	KV-Cache + Eviction	P8·P5 + X2	G	Dynamic addressing
A16	KV-Cache Quantization	P2·P5·P7	M	Scale (P2), clip+round (P7), find absmax (P5)
A17	Prefix Caching	P7·P8 + X2	G	Cache hit = P7; eviction policy needs X2
A18	KV-Cache Compression	P1·P5·P7·P8·P11 + X2	G	Track attn scores (P1·P8), rank (P5), evict (P7+X2)
A19	Prompt Caching	P7·P8 + X2	G	Hash-match-retrieve; needs associative lookup
A20	Speculative Decoding	P5·P7·P10·P12 + X1·X3	U	Token comparison (X1) + variable-depth rollback (X3)

2 Activation Functions (B1–B12)

#	Algorithm	Factorization	St.	Note
B1	ReLU	P7·P11	M	Comparator + analog switch; or diode
B2	Leaky ReLU	P7·P11·P2·P1	M	Two scaled branches gated by P7, summed
B3	PReLU (parametric)	P7·P11·P2·P1	M	Inference = Leaky ReLU (learned α fixed)
B4	ELU	P7·P11·P3·P1·P2	M	Negative branch: $\alpha(e^x - 1)$ via P3·P1·P2
B5	Swish / SiLU	P11·P6	M	$x \cdot \sigma(x)$: Gilbert $\times$ FET-pair sigmoid
B6	Mish	P11·P6·P3·P1·P4	M	$x \cdot \tanh(\ln(1 + e^x))$; 5 primes deep, all physical
B7	SoftPlus	P4·P1·P3	M	$\ln(1+e^x)$: subthreshold FET $\rightarrow$ junction $\rightarrow$ diode
B8	GELU (approx)	P11·P6·P2	M	Approx $x \cdot \sigma(1.702x)$: same as Swish with P2 prescale
B9	Tanh	P6·P2·P1	M	$2\sigma(2x) - 1$: differential FET pair gives tanh directly
B10	HardSigmoid	P7·P1·P2	M	$\text{clip}(x/6 + 0.5, 0, 1)$: divider + dual comparator
B11	HardSwish	P11·P7·P1·P2	M	$x \cdot \text{HardSigmoid}(x)$: Gilbert $\times$ clipped-linear
B12	Squared ReLU	P11·P7	M	$\text{ReLU}(x)^2$: comparator + squarer

3 Normalization (C1–C7)

#	Algorithm	Factorization	St.	Note
C1	RMSNorm	P1·P2·P11·P12	M	Solved: AGC feedback or translinear pipeline (this paper)
C2	LayerNorm	P1·P2·P11·P12	M	RMSNorm + mean subtraction (P1·P2 for μ)
C3	BatchNorm (training)	P1·P2·P11·P12·P8 + X2	G	Cross-sample stats need dynamic routing (X2)
C4	BatchNorm (inference)	P1·P2	M	Frozen: fixed affine $\gamma(x-\mu)/\sigma + \beta$
C5	GroupNorm	P1·P2·P11·P12	M	Stats over fixed channel groups; no cross-sample dep.
C6	InstanceNorm	P1·P2·P11·P12	M	GroupNorm with group = 1 channel
C7	Weight Normalization	P1·P2·P11·P12	M	$w = g \cdot v/\ v\ $: norm (P1·P11·P12) + scale

4 Loss Functions (D1–D8)

#	Algorithm	Factorization	St.	Note
D1	Cross-Entropy Loss	P3·P4·P1·P2	M	Solved: race + diode + KCL
D2	MSE Loss	P1·P2·P11	M	Subtract, square (P11), sum, scale
D3	L1 Loss	P1·P2·P7·P11	M	Full-wave rectifier (P7·P11 for $ x $)
D4	Huber Loss	P1·P2·P7·P11	M	Piecewise: P7 selects quadratic vs linear branch
D5	Focal Loss	P1·P2·P3·P4·P6·P11	M	$(1 - p)^\gamma \log p$: power via P3·P2·P4
D6	Triplet Margin Loss	P1·P2·P7·P11·P12	M	Distances + margin + ReLU clamp
D7	KL Divergence	P1·P4·P11	M	$\sum p \ln(p/q)$: log-ratio via diodes = translinear
D8	CTC Loss	P1·P3·P4·P11 + X2·X3	U	Dynamic alignment graph + unbounded recursion

5 Pooling (E1–E4)

#	Algorithm	Factorization	St.	Note
E1	Average Pooling	P1·P2	M	Resistor averaging network
E2	Max Pooling	P5	M	WTA over fixed receptive field; 1 prime
E3	Adaptive Average Pooling	P1·P2 + X2	G	Dynamic window size; if fixed at compile time $\rightarrow$ M
E4	Global Average Pooling	P1·P2	M	AvgPool with window = full spatial extent

6 Convolutions (F1–F6)

#	Algorithm	Factorization	St.	Note
F1	Conv1D	P1·P11·P8	M	Crossbar with sliding window
F2	Conv2D	P1·P11·P8	M	2D crossbar tiling
F3	Depthwise Separable Conv	P1·P11·P8	M	Per-channel spatial + 1×1 pointwise
F4	Dilated / Atrous Conv	P1·P2·P11·P8	M	P2 scales stride spacing by dilation rate
F5	Transposed Conv	P1·P11·P8	M	Same primes as Conv2D; transposition = routing
F6	Winograd Conv	P1·P2·P11	M	Pre/post-transforms (P1·P2), element-wise (P11)

7 Optimizers (G1–G7)

#	Algorithm	Factorization	St.	Note
G1	SGD + Momentum	P1·P2·P8	M	Leaky integrator = capacitor + resistive drain
G2	Adam / AdamW	P1·P2·P11·P12·P8	M	3×param storage; impractical >1M params
G3	AdaGrad	P1·P2·P11·P12·P8	M	Accumulated squared gradients (P8·P11)
G4	RMSProp	P1·P2·P11·P12·P8	M	EMA of squared grads (leaky P8)
G5	LAMB / LARS	P1·P2·P11·P12·P8	M	Adam + layer-wise trust ratio (extra P12)
G6	LION	P1·P2·P7·P8	M	sign() = P7; no second moment → no P11, P12
G7	Gradient Clipping	P1·P2·P7·P11·P12	M	Norm (P1·P11), compare (P7), rescale (P12·P2)

8 Training Techniques (H1–H5)

#	Algorithm	Factorization	St.	Note
H1	Backpropagation	P11·P1·P9 + X4	G	Graph tracing (X4) on GPU via torch.compile
H2	Gradient Checkpointing	X4	G	Pure control flow
H3	Mixed Precision	P2	M	Variable scaling factors
H4	Knowledge Distillation	P1·P2·P3·P4 + X4	G	Inference mappable; training needs X4
H5	Contrastive Learning	P1·P2·P3·P11	M	Batch pairwise similarity; $O(B^2)$ matmul

9 Positional Encoding (I1–I3)

#	Algorithm	Factorization	St.	Note
I1	Sinusoidal (fixed)	P1·P2·P3·P4	M	LC oscillator bank at geometric frequencies
I2	ALiBi	P1·P2·P7	M	Linear ramp $-m i-j $; drastically simpler
I3	Learned Embeddings	P1·P8·P11	M	Lookup = one-hot $\times$ embedding matrix

10 Generative Models (J1–J5)

#	Algorithm	Factorization	St.	Note
J1	Diffusion Reverse	P3·P1·P2·P10·P11	M	T serial passes; solved: Ex-tropic DTCA
J2	Thermodynamic Denoising	P10·P7·P1·P2·P3·P8	M	Hardware BM chain; XTR-0 board built
J3	VAE Reparametrization	P1·P2·P10	M	Solved: DC offset + scaled noise
J4	GAN Training	P1·P2·P3·P6·P11 + X4	G	Inference M; training needs backprop
J5	Flow Matching	P1·P2·P11·P8	M	ODE Euler steps; all mappable

11 Sequence & State-Space Models (K1–K5)

#	Algorithm	Factorization	St.	Note
K1	S4D (time-invariant SSM)	P8·P2·P1	M	Solved: memristor CIM with native decay
K2	Mamba (selective SSM)	P8·P2·P1·P11 + X2	G	Input-dep. A,B,C reprogramming
K3	LSTM / GRU Gates	P6·P11·P1·P2·P8	M	3-gate timing coordination
K4	Thermo Sampling + MatInv	P10·P1·P2·P8	M	Normal Computing SPU/CN101
K5	Classifier-Free Guidance	P1·P2	M	Weighted sum of two passes; trivial

12 Reinforcement Learning (L1–L5)

#	Algorithm	Factorization	St.	Note
L1	PPO / TRPO	P1·P2·P4·P11·P12 + X4	G	Policy gradient needs backprop
L2	TD Learning	P1·P2·P8	M	Accumulation + scaling + delay
L3	Experience Replay	P8 + X2	G	Random access (X2 on GPU)
L4	RLHF Pipeline	P1·P2·P3·P6·P11 + X4·X2	G	Multi-model orchestration
L5	MCTS	P5·P10 + X2·X3	U	Unbounded dynamic tree

13 GNN & Graph (M1–M4)

#	Algorithm	Factorization	St.	Note
M1	Message Passing GNN	P1·P2·P11	M	Fixed graph topology = fixed crossbar
M2	GAT (Graph Attention)	P1·P2·P3·P5·P11	M	Softmax over neighbors; fixed graph
M3	WL-Test / Graph Iso	X1	U	Symbolic hashing
M4	Dynamic Graph Update	P1·P2 + X2	G	Topology change at runtime

14 Numerics & Scientific (N1–N4)

#	Algorithm	Factorization	St.	Note
N1	Sparse Matrix Ops	P1·P2	M	Area efficiency, not feasibility
N2	Iterative Solver (CG)	P1·P2·P11	M	Convergence check needs P7 or ADC
N3	SVD / Eigendecomp	P1·P2·P11	M	Power iteration analog; full SVD serial
N4	Auto Differentiation	X4	G	Symbolic tracing (X4 on GPU)

15 Quantization & Efficiency (O1–O5)

#	Algorithm	Factorization	St.	Note
O1	GPTQ	P1·P2·P7·P11·P12·P8	M	Hessian inverse (P12), column-wise round (P7)
O2	AWQ	P1·P2·P7·P11·P8	M	Activation-aware scaling + quantization
O3	SmoothQuant	P1·P2·P7·P11	M	Per-channel scaling migration
O4	LoRA	P1·P11·P8	M	$W + BA$: two small matmuls + sum
O5	QLoRA	P1·P2·P7·P11·P8	M	LoRA + NF4 quantization of frozen W

16 Sampling & Decoding (P1–P5)

#	Algorithm	Factorization	St.	Note
P1*	Top-K Sampling	P5·P3	M	Solved: K-fold race
P2*	Nucleus (Top-p) Sampling	P1·P2·P3·P5·P7·P10·P11	M	CumSum + threshold + sample from remaining
P3*	Temperature Scaling	P2·P12	M	z/T : voltage divider. Minimal factorization
P4*	Repetition Penalty	P1·P2·P7·P8·P11 + X2	G	Variable-length token history needs X2
P5*	Beam Search	P5·P4 + X2·X3	U	Dynamic hypothesis tree

*IDs prefixed to avoid collision with prime notation.

17 Parallelism & Infrastructure (Q1–Q5)

#	Algorithm	Factorization	St.	Note
Q1	Tensor Parallelism	P1·P11·P8 + X2	G	Partial matmul + all-reduce (P1 across network)
Q2	Pipeline Parallelism	P1·P11·P8 + X2	G	Standard layer primes; schedule needs X2
Q3	Ring Attention	P11·P2·P3·P1·P8 + X2	G	Chunked attn with ring comm pattern
Q4	MoE Routing	P5·P1·P2·P3	M	Top-K race; bias = load balancing
Q5	Mixture-of-Depths	P5·P6·P7·P11 + X2	G	Router scores → top-k skip; dynamic per token

18 Probabilistic & Misc (R1–R4)

#	Algorithm	Factorization	St.	Note
R1	Bayesian NN Inference	P10·P1·P2·P3·P11	M	MCMC via thermal sampling
R2	BPE Tokenizer	X1	U	Symbolic; no continuous structure
R3	Dropout	P7·P10	M	Random mask (P10) × threshold gate (P7)
R4	Stochastic Depth	P7·P10	M	Random layer skip; same primes as Dropout

19 Summary Statistics (recounted in version 3)

The previous version’s summary line read 79 M / 22 G / 6 U over a stated total of 107. That line did not agree with the table above. Counting the rows of the eighteen tables of this document one by one gives **114** rows—A1–A20, B1–B12, C1–C7, D1–D8, E1–E4, F1–F6, G1–G7, H1–H5, I1–I3, J1–J5, K1–K5, L1–L5, M1–M4, N1–N4, O1–O5, P1–P5, Q1–Q5, R1–R4, with no duplicate identifier and no row carrying more than one algorithm—of which 86 are marked **M**. The M column and the total were understated by seven; the G and U columns were correct.

Status (v2 reading)	Count	Fraction
M (fully mappable)	86	75.4%
G (GPU-mappable)	22	19.3%
U (unmappable)	6	5.3%
Total	114	100%

Under the version 3 reading of the missing primes, the 28 non-M entries are re-derived: seven rows (A4, H2, K2, Q1, Q2, Q3, Q5) become plain **M**, one (R2) becomes **D** (digital-by-position), and the remaining twenty become **B** (bounded-mappable, with the bound named). The revised counts are:

Status (v3 reading)	Count	Fraction
M (fully mappable)	93	81.6%
B (bounded-mappable)	20	17.5%
D (digital-by-position)	1	0.9%
U (fundamentally unmappable)	0	0%
Total	114	100%

Key findings:

- 86 of 114 operations (75.4%) are fully realizable on analog hardware using only P1–P12 under the v2 reading; 93 of 114 (81.6%) under the v3 reading.
- All 22 GPU-mappable operations are blocked by X2 (dynamic topology) or X4 (symbolic tracing), both available on modern GPUs.
- Under the v2 reading, 6 operations (5.3%) were called fundamentally unmappable: Speculative Decoding (X1+X3), MCTS (X2+X3), Beam Search (X2+X3), CTC Loss (X2+X3), WL-Test (X1), BPE Tokenizer (X1). The main paper re-derives all six as bounded-mappable or digital-by-position.
- Recounting the factorization column: the most frequently invoked primes are P1 (Σ , 88 appearances), P2 (scaling, 78) and P11 ($\times$, 72) — accumulation, scaling and variable multiplication are the backbone of neural computation. (The previous version’s “89” and “62” do not reproduce.)
- The minimal factorization among entries with an explicit prime product is Temperature Scaling (P2·P12, 2 primes); four entries (H2, M3, N4, R2) carry no physical prime at all in the v2 table. The maximal is 7 primes, reached by three entries: FlashAttention, Sparse Attention (BigBird) and Nucleus Sampling.